

The non-parametric/machine learning future?

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Nonparametrics?

What do you mean by non-parametric?

Mostly we mean putting a flexible distribution on $f(\beta_i, \alpha_i \mid \theta)$ (and keeping logit error on ε_{ij})

$$u_{ij} = \beta_i x_j - \alpha_i p_j + \xi_j + \varepsilon_{ij} \text{ with } f(\beta_i, \alpha_i \mid \theta)$$

- Fixed Grids (Fox, Kim, Ryan, Bajari 2011, Heiss, Hetzenecker, Osterhaus 2022, Nevo Turner Williams 2016; see the BLP lecture): draw from “prior” of β_i , compute $\sigma_{ij}(\beta_i)$ and choose weights on each i, w_i .
- Compiani (QE 2022): approximate $\sigma_j^{-1}(\mathbf{s}_t, \mathbf{x}_t^{(2)})$ directly with Bernstein Polynomials (ditches $\varepsilon \rightarrow$ very hard)
- Ao Wang (JE 2022): use polynomial sieves: $\mathbb{E} \left[\left(\sigma_j^{-1}(\mathbf{s}_t; \mathbf{x}_t^{(2)}, F) - X_t^{(1)} \beta^{(1)} \right) \phi_k(Z_{jt}) \right] = 0$
- Lu, Shi, Tao (JE 2023): use partially linear model: $\log(s_{jt}/s_{0t}) = X'_{1,jt} \beta^0 + \psi^0(X_{2,jt}; IV_{J,t}) + \xi_{jt}$

$$\text{where } \psi^0(x_{2,jt}; IV_{J,t}) = \log \left[\frac{\int \frac{\exp(x'_{2,jt} v)}{\exp(IV_{J,t}(v))} f^0(v) dv}{\int \frac{1}{\exp(IV_{J,t}(v))} f^0(v) dv} \right].$$

But these are still **only as good as characteristics**.

Idea #0: Learn the Inverse Demand Directly (Nevo, Tao, Gandhi 2026)

Berry (1994) logic, generalized: for $u_{ijt} = x_{jt}\beta + \xi_{jt} + x_{jt}^{(2)}\nu_i + \varepsilon_{ijt}$ with iid ε_{ijt} and **any** $F_0(\nu)$, the inverse demand is

$$\sigma_j^{-1}(\mathbf{s}_t, \mathbf{x}_t^{(2)}) = \log(s_{jt}/s_{0t}) - h_0(\boldsymbol{\omega}_{jt}), \quad \boldsymbol{\omega}_{jt} = \{(s_{kt}, x_{kt}^{(2)} - x_{jt}^{(2)}, \mathbb{1}[k=0])\}_{k \neq j},$$

where h_0 is the **same function for every j** and **symmetric** (permutation invariant) in the other products k .

- Logit: $h_0 = 0$. Nested logit: $h_0 = \rho \log(s_{jt}/s_{G(j)t})$, a symmetric function of the shares in your nest. In general h_0 corrects the log odds for **how crowded the neighborhood of j is**: a high share means high quality *or* no close substitutes.
- Symmetric functions have a sum decomposition (**Deep Sets**, Zaheer et al. 2017): $h_0(\boldsymbol{\omega}_{jt}) \approx \psi(\sum_{k \neq j} \phi(\boldsymbol{\omega}_{jkt}))$. Their economic version, $\psi(\sum_{k \neq j} s_{kt} \phi(x_{kt}^{(2)} - x_{jt}^{(2)}, \mathbb{1}[k=0]))$: ϕ learns a **distance** in characteristic space, the sum adds up share within each distance band, ψ learns how to weight the bands. Nested logit with learned nests.
- ϕ and ψ are small neural nets (or symmetric polynomial sieves); the parameter count does not grow with J and evaluation is linear in J .

Deep Sets Demand: Estimation and How It Does

Step 1 (linear IV, no contraction): $\mathbb{E}[\xi_{jt} \mid z_t] = 0$ with $\xi_{jt} = \log(s_{jt}/s_{0t}) - x_{jt}\beta - h(\omega_{jt})$. Simplest implementation: predict $\hat{s}_{kt}$ from z_t , plug into the sum, 2SLS for β and the top layer. Own and cross elasticities from the IFT on the fitted inverse (as in Compiani 2022).

Step 2 (optional, needs T1EV): with $\hat{\delta}_{jt}$ in hand, recover $F_0(\nu)$ on a fixed grid of types by ridge-penalized nonnegative least squares of s_{jt} on type shares (FKRB; Heiss et al. 2022).

- ▶ Monte Carlo ($T = 250$, $J = 10$ to 20): when F_0 is normal, BLP wins and the flexible estimator loses little. When F_0 is bimodal, normal-RC BLP has own-elasticity bias around 15% and cross around 50%; the two-step estimator is around 1% on own with a fraction of the RMSE, and within-cluster merger simulations have RMSE 2.5 vs. 7.3 percentage points.
- ▶ What drives it: how well step 1 recovers δ_{jt} (correlation 0.993 vs. 0.9999 for correctly specified BLP). Cross elasticities and the tails are the hard part for everyone.
- ▶ Relatives: Compiani (2022), Monardo (2026), Lu Shi Tao (2023), Singh Liu Yoganarasimhan (2024, neural nets), Gandhi Houde (2025, same symmetry result).
- ▶ The catch from the first slide: symmetry fails when horizontal differentiation is **not in** $x^{(2)}$ (ε no longer iid). So: where do characteristics come from?

Idea #1: t-STE Embeddings (Van Der Maaten and Weinberger, 2012)

- **Goal:** Assign a low-dimensional vector of characteristics $x_j \in \mathbb{R}^m$ for each product using triplets of the form “ j is more similar to k than to ℓ ”.
- The “t” in t-STE comes from using a **Student-t kernel** to model distances—giving heavy tails and better separation of clusters.

$$\max_{\mathbf{x}} \sum_{(j,k,\ell) \in \mathcal{T}} \ln(\pi_{jkl}) \quad \text{where} \quad \pi_{jkl} = \frac{\left(1 + \frac{\|x_j - x_\ell\|^2}{\alpha}\right)^{-\frac{\alpha+1}{2}}}{\left(1 + \frac{\|x_j - x_k\|^2}{\alpha}\right)^{-\frac{\alpha+1}{2}} + \left(1 + \frac{\|x_j - x_\ell\|^2}{\alpha}\right)^{-\frac{\alpha+1}{2}}}$$

- This is basically maximum (log) likelihood for a binary outcome model (closer to ℓ than k).
 - Fit with canned routine (which does gradient descent).
 - α is a (somewhat arbitrary) tuning parameter.

Unobserved Characteristics: Magnolfi Maclure Sorensen (2023)

FIGURE 1: Sample survey page

Which two cereals on the right are most similar to FROOT LOOPS?

	SPECIAL K VANILLA ALMOND 	KIX 	CINNAMON LIFE 	FIBER ONE HONEY CLUSTERS 
	FRUITY CHEERIOS 	FROSTED FLAKES 	COOKIE-CRISP 	COCOA PUFFS 
	☐	☐	☐	☐

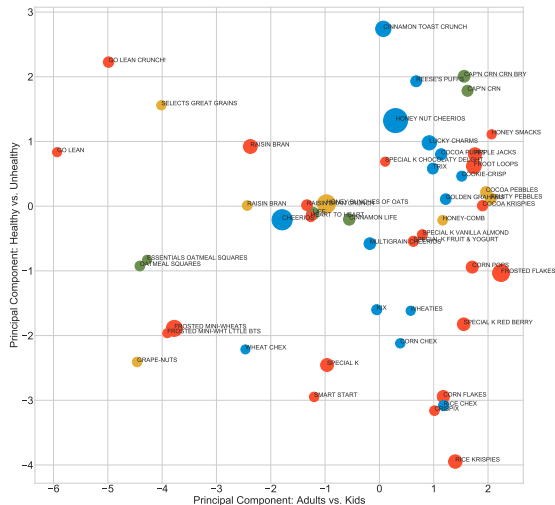
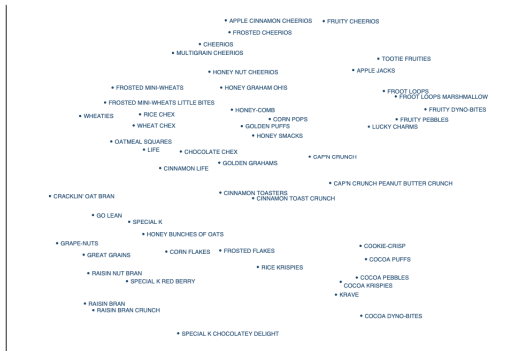
Submit

What if we could first estimate **unobserved characteristics**?

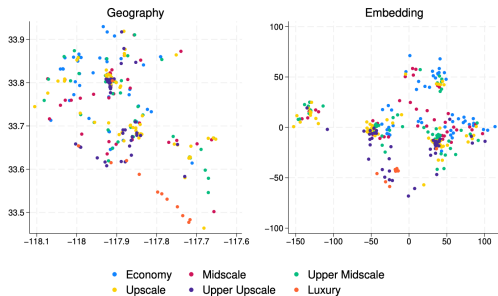
- Is j more similar to k or l ?
- Get a $m \times J$ matrix with m factors (embeddings).
- Idea: m is small (like 3-4).
- Use these as characteristics in BLP demand model.

Unobserved Characteristics: Magnolfi McClure Sorensen (2023)

FIGURE 2: Plot of two-dimensional embedding



Reverse Engineering Hotel Recommendations: McClure (2025)



APPENDIX FIGURE 1: Recommendations at Booking.com

Travelers who viewed Marriott Marquis Chicago ended up booking these properties

Show more



Hampton Inn Chicago McCor...

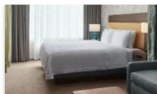
8.3 Very Good

📍 2 miles from center

♻️ Travel Sustainable property

Starting from

\$458



Home2 Suites By Hilton Chica...

8.4 Very Good

📍 2 miles from center

♻️ Travel Sustainable property

Starting from

\$441



Luxéry Stay Chicago - ACROS...

9.2 Wonderful

📍 1.9 miles from center

Starting from

\$500



Hilton Chicago

8.1 Very Good

📍 0.7 miles from center

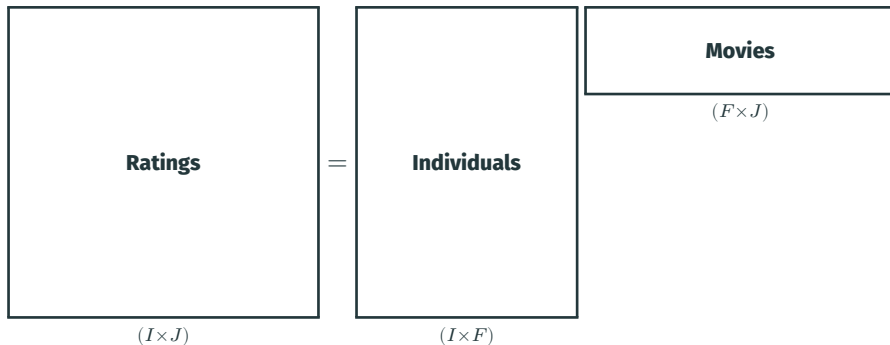
♻️ Travel Sustainable property

Starting from

\$544

- Scrape observed “you may also like” recommendations from hotels and use those to construct **triplets**
- Feed the triplets into tSTE embedding model to get a matrix $\mathbf{X}$: $(J \times F)$ where F is small (2-3).
- In both cases plug these in as characteristics to demand model (no guarantee they explain substitution patterns).

Idea #2: Low Rank Matrix Factorization: aka Netflix Prize



- ▶ Even if Ratings are **sparse**, we fit the observed cells and predict the rest!
- ▶ Idea: Approximate with a low rank (F) factor model (such as SVD)
- ▶ Maybe embeddings are reverse-engineering “collaborative filter”.

Second Choices as a Matrix: How Much Rank Do You Need?

Recall the second-choice decomposition from the diversion lecture; stacking it over (j, k) gives a matrix:

$$D_{j \rightarrow k} = \sum_{i=1}^I w_i \cdot \frac{\sigma_{ij}}{1 - \sigma_{ij}} \cdot \frac{\sigma_{ik}}{s_j} \iff \mathbf{D} = \text{diag}(\mathbf{s})^{-1} \tilde{\mathbf{S}} \text{diag}(\mathbf{w}) \mathbf{S}^\top, \quad \tilde{\sigma}_{ij} = \frac{\sigma_{ij}}{1 - \sigma_{ij}},$$

where the columns of $\mathbf{S}$ are the type-level choice-probability vectors σ_i .

Model	rank(D)	Why
Logit ($I = 1$)	≤ 1	one outer product $\tilde{\mathbf{s}} \mathbf{s}^\top$: proportional substitution
Nested logit (G nests)	$\leq G$	one outer product per (non-singleton) nest
Mixed logit (I types)	$\leq \min\{I, J + 1\}$	one rank-one term per type

- ▶ Logit and nested-logit bounds are **tight**; the mixed-logit bound is usually slack (thousands of draws on a handful of characteristics).
- ▶ Idea: skip characteristics and parametric random coefficients; **control rank(D) directly** in product space.

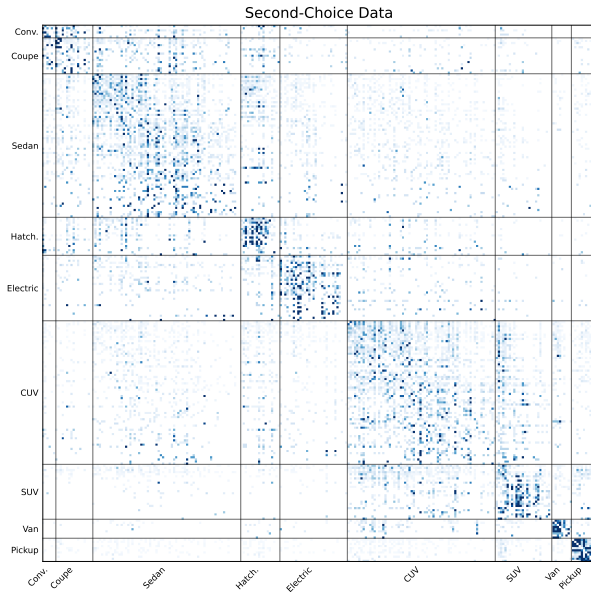
Conlon, Mortimer, Sarkis (2026): Second Choices in Product Space

Data from a **single market**: shares s_j and a subset Ω of survey second-choice cells $\mathcal{D}_{j \rightarrow k}$. For each rank I , solve

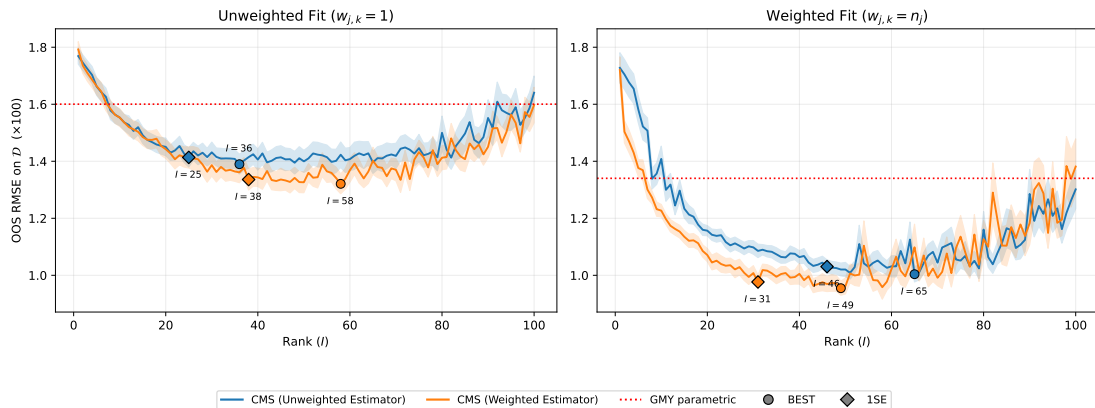
$$\min_{(\mathbf{S}, w) \geq 0} \sum_{(j,k) \in \Omega} \omega_{jk} (\mathcal{D}_{j \rightarrow k} - D_{j \rightarrow k}(\mathbf{S}, w))^2 + \lambda \|\mathbf{s} - \mathbf{S} w\|_{\ell_2}^2 \quad \text{s.t.} \quad \sum_j \sigma_{ij} = 1 \quad \forall i, \quad \sum_i w_i = 1.$$

- **Estimates** σ_i (type profiles) and w_i (type weights): a finite mixture with $I \ll J$ and no characteristics.
- **Tuning**: λ is a share-fit penalty (not a ridge); (I, λ) chosen by cross-validation on held-out cells of Ω .
- **Weights** $\omega_{jk} \propto n_j$, the respondents behind row j (approximately inverse-variance under row-multinomial sampling).
- **Computation**: reparameterize $\sigma_{ij} = e^{V_{ij}} / \sum_k e^{V_{ik}}, w_i = e^{\gamma_i} / \sum_\iota e^{\gamma_\iota}$, so the problem is unconstrained; autodiff + Adam. Optional sparsification zeroes $\geq 75\%$ of $\mathbf{S}$ (stop when AIC rises).
- **Matrix completion**: the fitted $\mathbf{D}(\mathbf{S}, w)$ predicts the *unobserved* cells $(j, k) \notin \Omega$.
- Software: [PyCMS](#) (PyTorch).

What the Data Look Like: MaritzCX Second Choices, 254 Cars and Light Trucks (2015)



Cross Validation: Out-of-Sample Fit



Held-out second-choice cells. Parametric BLP-style benchmark from Grieco, Murry, Yurukoglu (QJE 2023): $R^2 = 0.18$; CMS $I = 38$ (one-SE rule): 0.65; CMS $I = 58$ (best): 0.78.

Top Substitutes: Chevrolet Suburban (second-choice %)

Model	Raw	GMV (Parametric)	CMS 1SE (I=38)	CMS Best (I=58)
Gmc Yukon Xl	41.97	8.36	40.81	42.07
Ford Expedition	25.18	5.47	24.51	23.51
Toyota Sequoia	4.38	0.56	5.27	2.41
Chevrolet Tahoe	4.38	8.36	4.21	4.31
Cadillac Escalade	3.28	0.85	4.13	3.61
Gmc Yukon	2.19	2.99	2.83	4.42
Lincoln Navigator	1.82	0.69	5.73	5.15
Infiniti Qx80	1.46	0.48	0.47	0.74
Dodge Durango	1.09	2.20	0.73	0.79
Jeep Grand Cherokee	1.09	3.58	0.19	0.35

Raw survey vs. the parametric GMV model vs. CMS at two ranks. The parametric model spreads substitution across the segment; the data concentrate it on two near-twins.

Second Stage: Putting Prices Back In

The first stage delivers $\hat{\mathbf{S}}, \hat{w}$ and hence $V_{ij} - V_{i0}$, but no price sensitivity and no welfare. Assume $V_{ij} = V_{ij}^* - \alpha_i p_j$ (constant within type, free across types) and calibrate $\alpha_i > 0$ from one of:

- **Own-price elasticities** (a quasi-experiment, another study, a simpler demand system):

$$\min_{\alpha > 0} \left\| \mathcal{E}_{jj} + \frac{p_j}{\hat{s}_j} \sum_{i=1}^I \alpha_i \hat{w}_i \hat{\sigma}_{ij} (1 - \hat{\sigma}_{ij}) \right\|_{\ell_2} \quad (\text{need } \geq I \text{ targets, or } \alpha_i = \alpha)$$

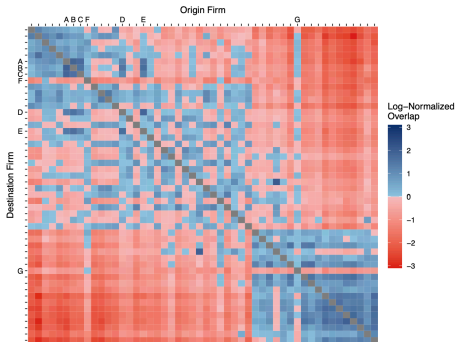
- **Price-cost margins**: Bertrand FOC with ownership matrix $\mathbf{H}$ and demand derivatives $\Delta(\alpha) = \sum_i \hat{w}_i \Delta_i(\alpha_i)$,

$$\mathbf{c}(\alpha) = \mathbf{p} - (\mathbf{H} \odot \Delta(\alpha))^{-1} \mathbf{s}, \quad \min_{\alpha > 0} \left\| \mathcal{C}_j - c_j(\alpha) \right\|_{\ell_2}.$$

- Either way you now have a full mixed logit in product space: elasticities, diversion under price changes, merger simulation (Week 7).
- **Inference**: each row of $\mathcal{D}$ is Multinomial($n_j, \cdot$); parametric bootstrap with warm-started re-solves. Type labels are not identified, so report CIs for **invariant functionals** only (fitted $D_{j \rightarrow k}$, elasticities). E.g. Suburban $\rightarrow$ Yukon XL + Expedition: 65%, 95% CI [60%, 71%].

Idea #3: Customer Overlap (Einav, Guido, Klenow, AER:J 2026)

Figure 4: Hotel Overlap



Note: This picture visualizes all pairwise log-normalized overlap measures for the top 50 hotel chains in the U.S. Hotel chains sorted from highest non-hotel spending rank to lowest non-hotel spending rank across both axes, going from left to right and from top to bottom (i.e. highest ranked chains are in the top left and lowest ranked chains are in the bottom right). A single cell of this matrix can be interpreted as the log-normalized overlap of the column chain with the row chain (i.e. $C_{col \rightarrow row}$). Thus, any column can be interpreted as how much that chain's customers relatively overlap with other chains. Similarly, any row can be interpreted as how much all other chains' customers overlap with the row chain. Values above zero are colored blue, with darker colors indicating higher log-normalized overlap. Similarly, values below zero are colored red, with darker shades

- ▶ Encode a $J \times 1$ vector $\mathbf{q}_i$ of every purchase (0/1) within the category for the year; count customers two sellers share.
- ▶ From the diversion lecture: the raw overlap $C_{j \rightarrow k}$ is quality diversion for short panels but collapses to IIA as $T_i \rightarrow \infty$; the odds-reweighted $A_{j \rightarrow k}$ converges to **second-choice diversion** instead.
- ▶ So panel data alone (credit cards: stores, not items) can supply the $\mathcal{D}$ matrix for the CMS program above, with no survey.
- ▶ Conlon Rao (JPE, forthcoming) use a similar measure to estimate the nesting parameter; Atalay et al. (JPE 2024) use one to assign products to nests.

Thanks!
